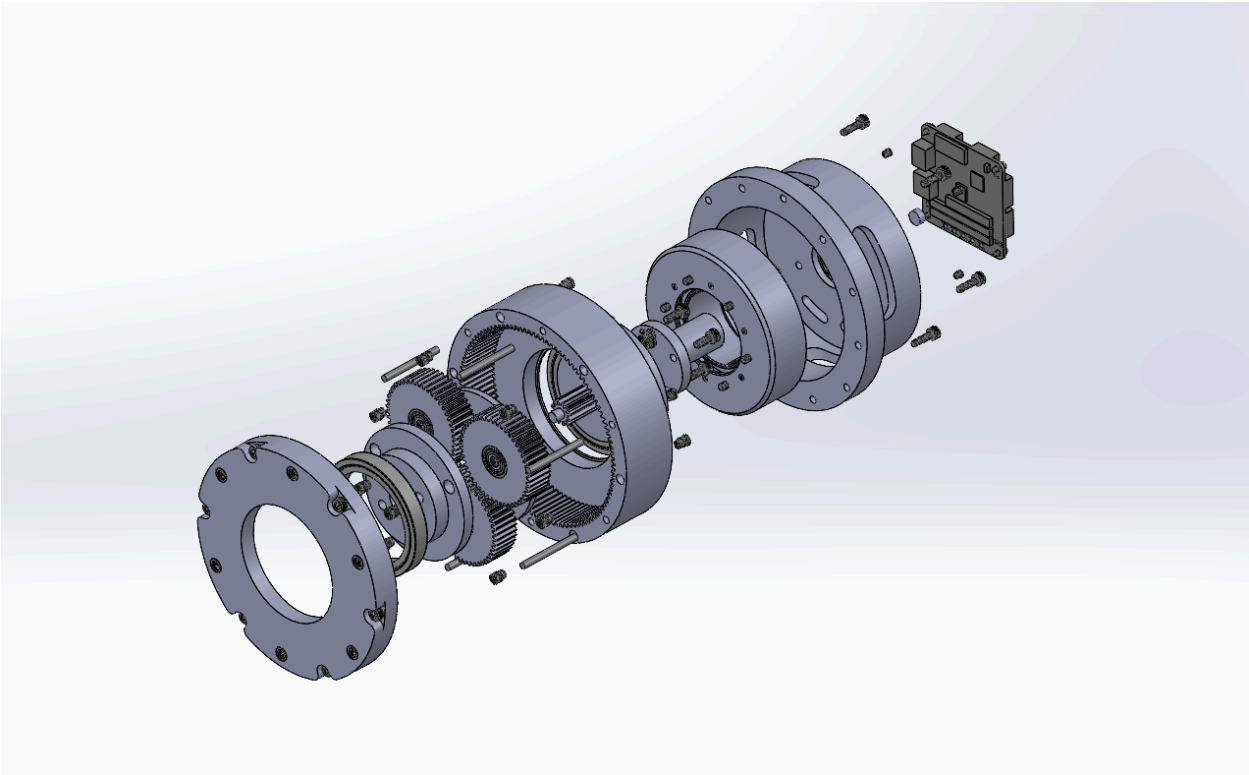


[1] - Full exploded view of assembly, showing encoder, motor housing, gearbox, input/output shaft, encoder interaction, bearings

Targeted Specifications

Specification description	Target	Unit	Test method	Rank
Performance				
Continuous (rated) Torque	6.6	Nm	Torque meter / dyne testing	1
Peak torque	19.8	Nm	Torque meter / dyne testing	1
Torque density	10.6	Nm/kg	Torque meter / dyne testing	2
Mass	<0.62	kg	Scale	2
Max Output speed at 0 load	40 at 48V	Rad/s	No load spin with encoder or tachometer at rated bus voltage	2
backlash	12	Arcminute	Precision angle gauge / runout	1
Backdrive torque	0.5	Nm	Torque meter / dyne testing	1
Continuous duty thermal limit	100	C	Thermocouples placed on	2

			actuator exterior housing	
Envelope dimensions		mm^3	N/A	3
Output Radial Load capacity	200	N	Load Cell	2
Output axial load capacity	100	N	Load Cell	2
Gear ratio	10:1	N/A	Torque meter / dyne testing	2
Gearbox Efficiency	90	%	Torque meter / dyne testing	2
BLDC torque (continuous)	0.75	Nm	Torque meter / dyne testing	1
BLDC torque (peak)	2.2	Nm	Torque meter / dyne testing	1
Electrical Power Input		Watts	Multimeter	2
Mechanical Power output		Watts	Torque meter / dyne testing combined with tachometer	2
Position repeatability/accuracy (bidirectional)	20	Arcminute	Precision angle gauge / runout	2
Cost and manufacturability				
Unit cost	200	$	Total the BOM and outsourced machining quotes	1
Domestic Sourcing	pass/fail	pass/fail	All parts should be sourced from US	2
Manufacturing processes	pass/fail	pass/fail	Only common shop methods (CNC, wire EDM, 3D printing)	2
Integration and setup				
Time from unboxing to first motion	15	min	Time a new user with only the docs	1
Standardized mounting interface	pass/fail	pass/fail	Fit check on a mating plate	2
Supply voltage range	24-48	VDC	Run the actuator across the range on a bench supply	1
Built in protection (overcurrent, overvoltage, overtemperature)	pass/fail	pass/fail	Trigger each fault on the bench and confirm the driver shuts off	1
Self tuning routing	0	Number of parameters user must set before first motion	N/A	1

Control and Security				
No cloud or internet connection required for setup and use	pass/fail		Will not include connectivity on board	1
Standard Digital Bus	pass/fail		Connect the actuator to an CAN FD or EtherCAT (TBD) master, Scan the network and confirm the actuator is detected and reaches the operational state. Send position, speed, and torque commands through process data and confirm the actuator responds and reports feedback.	1
Backdriving protection	pass/fail		Actuator must be safely backdriven and not cause electrical damage when backdriven	1